

SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

Packaging Machinery

Shrink Packaging Machines

01 Description

SJFM-2100 serial healthy environmental shower membrane machine and paper plastic composition unit (non-adhesive composition unit) which is integrated winding, drying, surface process, composition, edge-cutting, absorption edge and rewinding are adopted with photoelectric tracking, electric spark treatment, linear speed and length of material, digital display, full automatic eternal tension, suction automatic control etc advance technologies.

02 Features

(I) Technical Parameter

1. Applicable coating film resin: LDPE, PP, EVA, EAA ETC
2. Based material: paper 35~425g/m², fabrics, films, and other composite material
3. Max working speed: 80~150m/min
4. Max width of coating film: 2050mm
5. Thickness of coating film: 0.01~0.11mm
6. Inaccuracy thickness of coating film: ±6%
7. Setting range of Auto-tension : 10-200kg/entire width (constant tension)
8. Max extrusion: 260kg/h
9. Compound cooling roller: Φ600×2100mm
10. Total power: about 170kw
11. Max rewinding diameter: Φ1300mm
12. Inside diameter of base material: Φ76
13. Machine total weight: about 20000kg
14. Overall dimension: 9600mm×12000×3250mm (L×W×H)
15. Machine color: can choose



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



30 TAHUN
MELAYANI INDONESIA
Anniversary 1982-2012

SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

02 Features Continued

(II) Main equipment

1. Feeding equipment

(Structure)

(1) Double work-station bearing feeding frame

(2) Air expansion feeding shaft

(Specification)

(1) Effective width: 2050mm

(2) Max feeding diameter: $\Phi 1300\text{mm}$

(3) Inside diameter of base material : 3 inch

(4) Max weight air-expansion shaft support: 2000kg

(5) Tension setting: 10-200kg

(6) Tension control accuracy: $\pm 0.2\text{kg}$

(7) Magnetic powder clutch: 2

(8) Automatic tension control: 1

(9) Air-expansion feeding shaft: 3

(Characteristic)

1. Tension controller: you can input diameter and thickness parameter of base material based on replaced material, along with the rotated-loops changing; tension is decreased proportionally to achieve automatic tension control.

2. Corona Treater

Electric spark power: 6KW Corona Treater adopts dustproof ,interference-proof structure, pneumatic switch cover equipment, achieve discharging ozone.

2. Extrusion and compounding equipment

(Structure)

(1) Three roller compounding mechanism, back pressing roller makes compounding roller to force evenly and compounding firmly.

(2) Peeling off silicone roller: compounding products peels off from cooling roller easily.

(3) Preheating inside before compounding to guarantee fastness and intensity of film compounding effectively.

(4) Compounding and filling regulating roller can overcome defects such as uneven of film thickness.

(5) Die slitting knife edge-cutter: sharp knife edge

(6) High pressure blower sucks scrap edge quickly.



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



30 TAHUN
MELAYANI INDONESIA
Anniversary 1982-2012

SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

02 Features Continued

- (7) Compounding and discharging eliciting roller.
- (8) Compounding roller be driven by transducing electric motor independently.
- (9) The motor drived compounding roller is controlled by frequency changer.
- (10) Rewinding equipment adopts cylinder to switch automatically.

(Specifications)

- (1)Compounding roller: $\Phi 500\text{mm} \times 2100\text{mm}$ 1
- (2)Silicone roller: $\Phi 260 \times 2100\text{mm}$ 1
- (3)Back pressing roller: $\Phi 220 \times 2100\text{mm}$ 1
- (4) 15kw planetary speed reducer, motor 1
- (5)15kw frequency converter 1
- (6)Rotating joint 6

(Characteristic):

- (1) Cooling roller adopts high degree of finish roller, which can eliminate bubbles produced during the compounding process.
- (2) Silicone roller and cooling roller adopt screw-type cooling structure, to make cooling quickly and laminating easily.
- (3) Compounding roller is driven by one vector frequency electric motor

3. Extrusion equipment

- (1) Car type extrusion machine
- (2) T type die head
- (3) Automatic PE feeding equipment
- (4) Oil hydraulic pump changes filter automatically.
- (5) Extrusion machine can move back and forth, up and down.
- (6) Screw and charging barrel linkage area are all heated with cast aluminum board.
- (7) High power and helical gear speed reducer
- (8) Temperature is controlled by Digital temperature controller automatically.
- (9) Stainless steel hopper
- (10) Six screw and charging barrel heating zones are controlled independently.
- (11) Eleven die head heating zones are controlled independently



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



30 TAHUN
MELAYANI INDONESIA
Anniversary 1982-2012

SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

02 Features Continued

(Specification):

- (1) Effective width of die head , 2200mm; Clothes-hanger type runner, width of laminating,2050mm,adustable;
- (2) Screw diameter: $\Phi 110$
- (3) Ratio of screw length to diameter: 32:1
- (4) 55kw AC motor 1
- (5) 1.5kw Extrusion machine moving motor 1

(Characteristic):

- (1) Clothes-hanger type runner, the critical position(die-lip) is ground precisely after electroplating to make laminating
- (2) Big length-diameter ratio makes resin plasticizing more thoroughly, for better rewinding.

4. Rewinding equipment

(Structure):

1. Surface rewinding (Double work-station ,Non-stopping)
2. Surface rewinding roller is driven independently
3. Drive motor
4. Frequency speed control (shenzhen Inovance)
5. Rewinding shaft be pressed by compacting cylinder
6. Hydraulic pressure edge regulating

(Specification):

1. Rewinding diameter: $\Phi 1300\text{mm}$
2. Inside diameter of base material : 3inch
3. Width of film coating:2050mm
4. Surface rewinding roller, high degree of accuracy chrome Plated roller $\Phi 510$
5. 15kw planetary speed reducer ,motor
6. 15kw frequency converter (shenzhen Inovance)
7. Rewinding shaft press compacting cylinder: $\Phi 80\text{mm}$
8. Air-swollen feeding shaft



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



30 TAHUN
MELAYANI INDONESIA
Anniversary 1982-2012

SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

02 Features Continued

5. Control unit

- Independent control box control, temperature indication control
- PLC Center control

Character:

- Whole machine adopts PLC center control, Taiwan famous brand touch screen control
- Can simultaneously control the speed of the host and complex machine, the speed of the main machine speed increased while complex machine speed increase, decreased, proportional to the ascending or descending, keep the same lamination thickness.
- Can set the thickness of the lamination according to need, the host automatically with the complex machine speed the lamination thickness reaches the set.
- Multi-parameter memory function, the parameters stored in the system can be invoke at any time
- Tension, speed and work status **dynamic display**

6. Others

- Frequency control of motor speed
- Cylinder floating buffer pendulum roller tension detection precision positioner feedback.
- Rewinding shaft air clasper
- Other equipment
- Guide roller: $\varnothing 80$ — 120×2100 mm
- The rack adopts the wallboard structure
- Pneumatic: all are excellent quality, gas circuit can be controlled in the first touch.
- Cylinder (Taiwan)
- Low-voltage components: all are UC Certified products.
- Antriebseinheit
- Automatic mechanical transmission system (main motor, composite, the winding motor)

7. Guide roller: $\varnothing 120 \sim 90 \times 2100$ mm

8. Pneumatic: all are excellent quality; gas circuit can be controlled in the first touch.



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



SJFM-2100

High Speed Extrusion Laminating Composite Machine With Second Unwind

02 Features Continued

9. Low-voltage components: all are UC Certified products.

- Cylinder (Airtac Taiwan)
- Circuit breaker, Schneider French
- AC contactor, Schneider French
- 24Vauxiliary relay Schneider French
- 220V/N auxiliary relay, Japan Omron
- Button with lamp, Knob indicator, mushroom head button, ONPOW
- Switching power supply (Taiwan Meanwell, Solid-State Relay (zhejiang mager)
- Control box use simens brand

10. Whole machine is touch-sensitive screen man-machine interface.

11. Supporting facilities

1. Power: 3 phase 380V 50Hz (three phase four wire system)
2. Barometric pressure: 6~8/kg/cm²
3. Water pressure: 2~3kg/cm²

03 Technical Specification

Model	SJFM-2100
Film coating resin	LDPE, PP, EVA, EAA, etc
Paper thickness	35-425 g/m2
Max speed	80-150 m/min
Max width of coating film	2050 mm
Film coated thickness	0.01-0.11mm
Film coated thickness error	±6%
Tension setting scope	10-200kg/entire width, constant tension
Max. extrusion capacity	260 kg/h
Total Power	About 170kw
Basic material rolling core ID	Φ76mm / 3"
Weight	20000kg
Overall dimension	9600 x 12000 x 3250 mm



PT. EKA MAJU MESININDO
Jl. Terusan Bandengan Utara No.16 / B 3-4
Jakarta, 14450, Indonesia
Tel. (021) 660 2665, 661 8255.
Fax. (021) 668 5874, 660 2657



30 TAHUN
MELAYANI INDONESIA
Anniversary 1982-2012